DM – Graph connectivity

NGUYEN Hoang Thach
nhthach@math.ac.vn

Outline

1. Undirected graph
   - Path and connectedness
   - Vertex connectivity and edge connectivity
2. Directed graph

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Path

Definition

Let $G = (V, E)$ be a graph. A path of length n from a vertex u to a vertex v is defined by a sequence of edges $e_1, e_2, \ldots, e_n$ such that $e_i = (x_{i-1}, x_i)$ and $x_0 = u, x_n = v$.

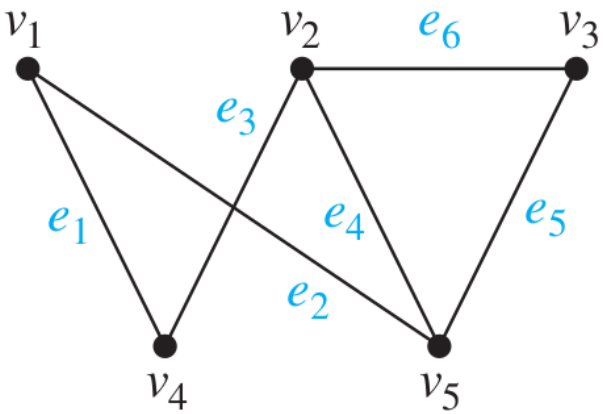
When the graph is simple, this path can be identified with the vertex sequence $u = x_0, x_1, x_2, \ldots, x_n = v$.

If $u = v$ and $n > 0$, then the path is a circuit (or cycle).

A path or circuit is simple if it does not pass by the same edge more than once.

Example:

- $e_1 e_2 e_4 e_6$ or $v_4 v_1 v_5 v_2 v_3$ is a path of length 4 from v_4 to v_3.
- $e_4 e_5 e_6$ is a circuit of length 3.



Distance and diameter

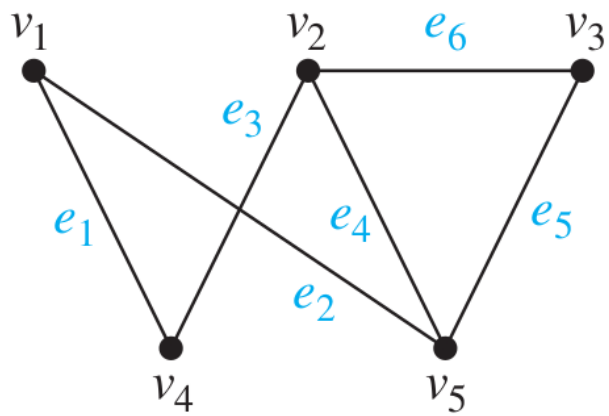
Definition

Let $G = (V, E)$ be a graph.

1. *The* distance *$d(u, v)$ between two vertices u and v is the length of the shortest* simple *path between them. If no such path exists, the distance is infinite.*
2. *The* diameter *of G is the greatest distance between any pair of its vertices.*

Example:

- $d(v_1, v_2) = 2\,, d(v_3, v_5) = 1.$
- The diameter of the graph is 2.

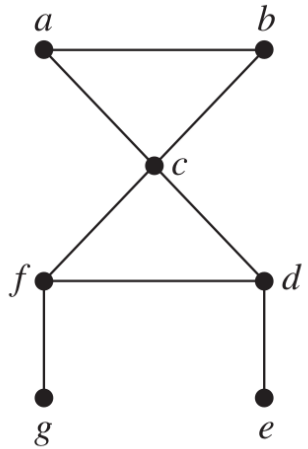


Connected graph

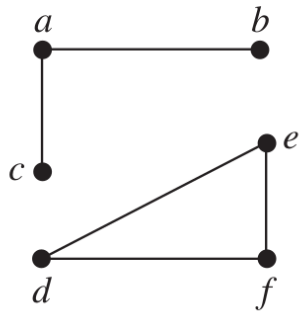
Definition

A graph G is connected *if there exists a path between every pair of its vertices.*

Examples: G_1 is connected, G_2 is not connected (or *disconnected*).



G_1



G_2

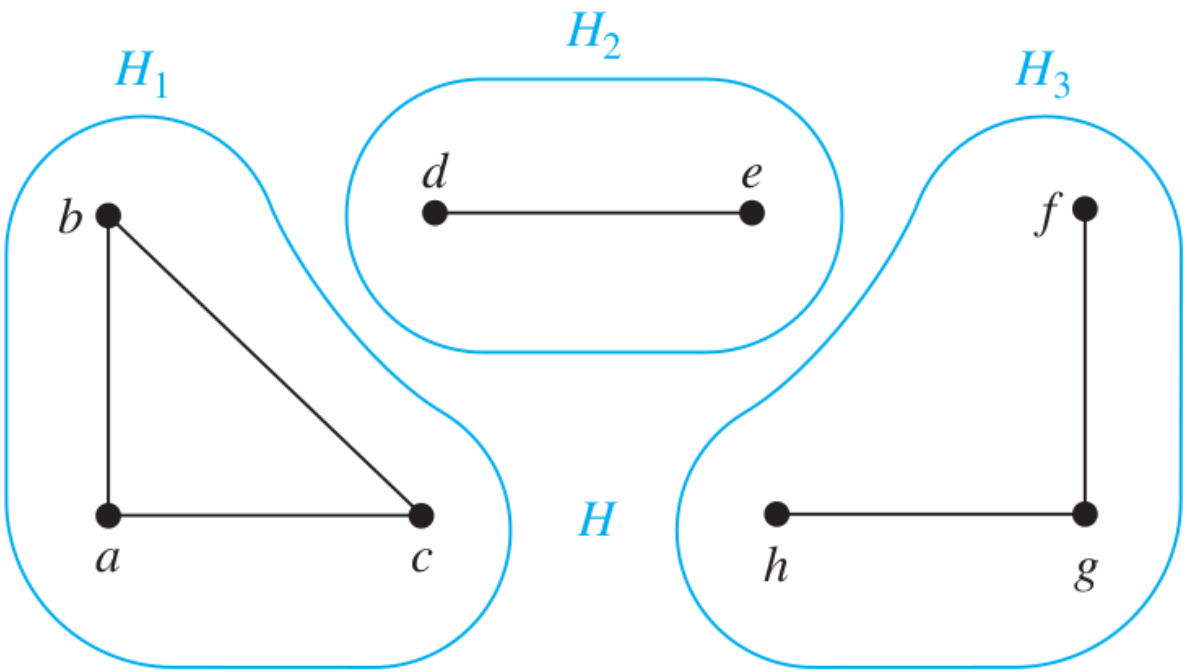
Theorem

There exists a simple *path between every pair of vertices of a connected graph.*

Connected component

Definition

Let G be a graph. A connected component *of G is a* maximal *connected subgraph of G, ie. it is not a* proper *subgraph of another connected subgraph of G.*



Counting paths

Theorem

Let G be a graph with adjacency matrix A. For all positive integer k, the number of different paths of length k from u to v is the (u, v)-th entry of A^k.

Note: The theorem also holds for multigraphs with loops.

Corollary

Let G be a graph with adjacency matrix A. The graph G is conneced if and only if there exists a positive integer k such that every entry of the matrix

$$B = \sum_{i=1}^{k} A^i$$

is positive.

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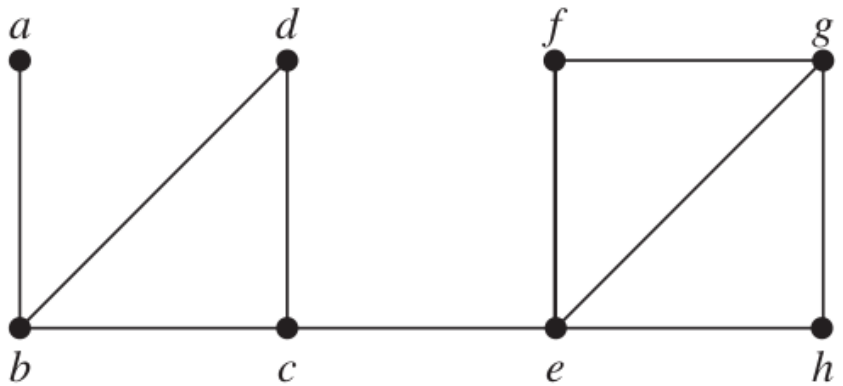
Cut vertices and bridges

Definition

Let G be a graph.

1. *A vertex v is a cut vertex (or articulation point) if $G - v$ has more connected components than G.*
2. *An edge e is a bridge if $G - e$ has more connected components than G.*

Examples: The vertex c is a cut vertex, the edge $\{c, e\}$ is a bridge. The vertex a is not a cut vertex.



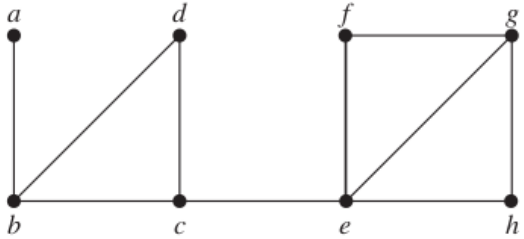
Vertex connectivity

Definition

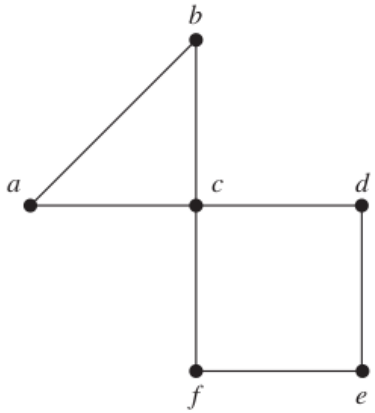
Let $G = (V, E)$ be a connected graph.

1. *A set $W \subset V$ is a vertex cut if $G - W$ is disconnected.*
2. *The vertex connectivity of G, denoted by $\kappa(G)$, is the size of the smallest vertex cut of G.*
   By convention, $\kappa(K_n) = n - 1$.
3. *The graph G is k-(vertex-)connected if $\kappa(G) \geq k$.*

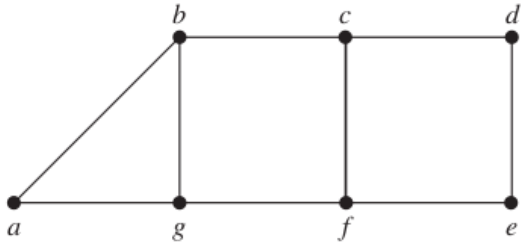
Vertex connectivity



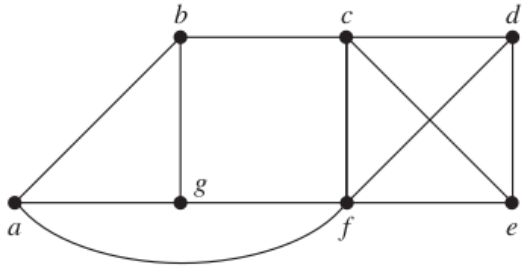
G_1



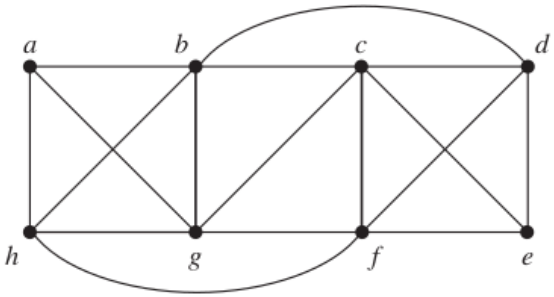
G_2



G_3



G_4



G_5

Edge connectivity

Definition

Let $G = (V, E)$ be a connected graph.

1. *A set $F \subset E$ is a edge cut if $G - F$ is disconnected.*
2. *The edge connectivity of G, denoted by $\lambda(G)$, is the size of the smallest edge cut of G.*
   By convention, $\lambda(K_1) = 0$.

Theorem

If $G = (V, E)$ is a non-complete connected graph with at least three vertices, then

$$\kappa(G) \leq \lambda(G) \leq \min_{v \in V} \deg(v)\,.$$

Path

Definition

Let $G = (V, A)$ be a directed graph. A path of length n from a vertex u to a vertex v is defined by a sequence of arcs $a_1, a_2, \ldots, a_n$ such that $a_i = (x_{i-1}, x_i)$ and $x_0 = u, x_n = v$.

When the directed graph is simple, this path can be identified with the vertex sequence $u = x_0, x_1, x_2, \ldots, x_n = v$.

If $u = v$ and $n > 0$, then the path is a circuit (or cycle).

A path or circuit is simple if it does not pass by the same edge more than once.

Connectedness

Definition

A directed graph G is strongly connected *if for every pair of vertices u and v, there is a path from u to v and a path from v to u.*

Note: A directed graph is *(weakly) connected* if the underlying undirected graph is connected. Strong connectedness implies weak connectedness, but the converse is not necessarily true.

Example: The graph G is strongly connected. The graph H is connected but not strongly connected (there is no path from a to b).

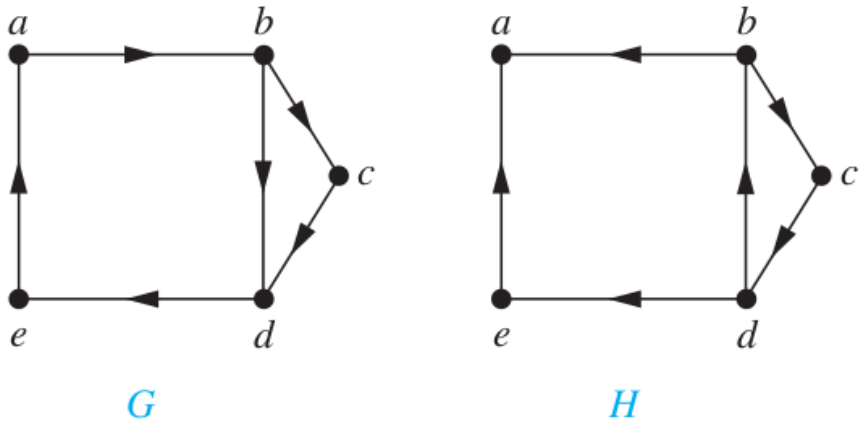


Figure: Rosen, p. 686

Strongly connected component

Definition

A strongly connected component of a directed graph G is a maximal strongly connected subgraph of G.

Example:

- The graph G has one strongly connected component which is itself.
- The graph H has three strongly connected components, which are the subgraphs induced by the sets of vertices $\{a\}, \{e\}, \{b, c, d\}$.

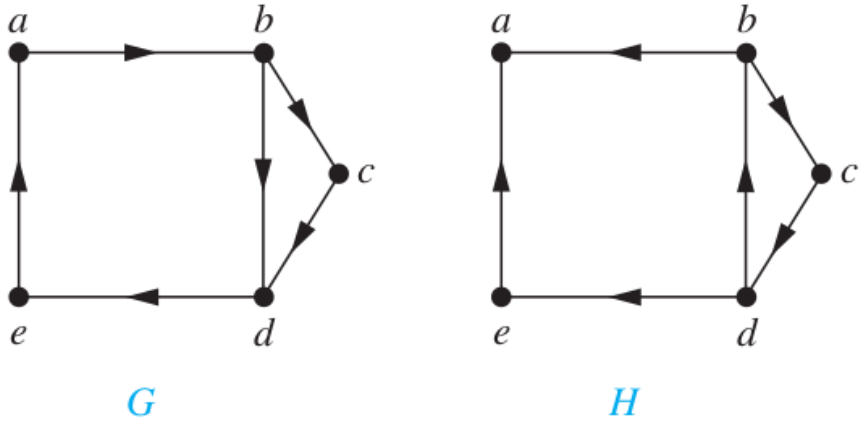


Figure: Rosen, p. 686